

Let f be a linear operator on the vector space $V = \text{Span}\{(2, 1, 0, 1)^T, (1, 0, 1, 2)^T\}$ such that

$$f((2, 1, 0, 2)^T) = (4, 1, 2, 5)^T, \quad f((1, 0, 1, 2)^T) = (2, 2, -2, -2)^T$$

Find an explicit expression for $f^n((5, 1, 3, 7)^T)$ with regard to n which doesn't use the symbol of the operator f .

Let B be the basis of V consisting of

$$B = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\}$$

From which we get

$$\left[\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \right]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \left[\begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \right]_B = \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \quad \left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Therefore

$$[f]_B^B \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad [f]_B^B \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

Or the matrix of the operator f in regard to the basis B is

$$[f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\det([f]_B^B - \lambda I) = \lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3) \implies \lambda_1 = 2 \quad \& \quad \lambda_2 = -3$$

Then

$$[f]_B^B \vec{v}_i = \lambda_i \vec{v}_i \iff \begin{pmatrix} 1 - \lambda_i & 2 \\ 2 & -2 - \lambda_i \end{pmatrix} \vec{v}_i = \vec{0}$$

$$\lambda_1 = 2 : \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 2 & -4 & 0 \end{array} \right) \approx \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \implies \vec{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = -3 : \left(\begin{array}{cc|c} 4 & 2 & 0 \\ 2 & 1 & 0 \end{array} \right) \approx \left(\begin{array}{cc|c} 4 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \implies \vec{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

From which we get

$$[f]_B^B = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 2/5 & 1/5 \\ -1/5 & 2/5 \end{pmatrix}$$

Therefore

$$\begin{aligned} \left[f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) \right]_B &= \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 2/5 & 1/5 \\ -1/5 & 2/5 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ \left[f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) \right]_B &= \frac{1}{5} \begin{pmatrix} (-3)^n + 10 \cdot 2^n - 6 \cdot (-3)^n \\ -2 \cdot (-3)^n + 5 \cdot 2^n + 12 \cdot (-3)^n \end{pmatrix} = \begin{pmatrix} 2 \cdot 2^n - (-3)^n \\ 2^n + 2 \cdot (-3)^n \end{pmatrix} \\ f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) &= \begin{pmatrix} 2 \cdot 2^n - (-3)^n \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 2^n + 2 \cdot (-3)^n \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \end{aligned}$$

Q. E. D.